

SIP-HAPUS: A Digitalisation and Optimisation System for Motor Vehicle Registration Deletion in Central Java.

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Abstract—The process of deleting motor vehicle registrations in Central Java involves multiple government agencies, including Samsat, Regional Police (Polda), Regional Revenue Agency (Bapenda), and PT Jasa Raharja. Cross-agency coordination relying on physical documents creates structural problems, including lengthy non-integrated processes, lack of transparency in monitoring status, and absence of centralized activity documentation. This aligns with the concept of *governance gap* in digital public service governance. This research aims to design and develop a web-based integrated monitoring platform for digitizing the motor vehicle registration deletion workflow, providing *real-time* status tracking, a monitoring dashboard, Decision Letter management, automated notifications, and *role-based access control*. The *Design Science Research* framework was employed with six stages. System development uses Laravel with Model-View-Controller architecture, MySQL database, Docker containerization, and Google Cloud Platform deployment. Evaluation uses expert heuristic evaluation based on Nielsen's 10 usability principles. The platform was developed with seven integrated modules. Heuristic evaluation showed 82.9% of items scored "Good." The System Usability Scale simulation yielded a score of 71.5 (Grade B), exceeding the e-government average of 68. The estimated task completion rate reached 92.3%. The SIP-HAPUS platform integrated motor vehicle tax cancellation processes into a transparent, real-time, and secure digital system, achieving a 96% time reduction

from 90–120 days (manual) to 2–4 days (digital).

Keywords: Registration Deletion, Design Science Research, Laravel, Heuristic Evaluation, Monitoring System, E-Government

I. INTRODUCTION

In Central Java, the process of deregistering motor vehicle registrations involves several agencies such as Samsat, Polda, Bapenda, and PT Jasa Raharja [1]. Coordination among different government agencies based on physical documents leads to some issues, namely long non-integrated process, lack of transparency in tracking status, and lack of central activity documentation. These issues are associated with the idea of *governance gap* in digital public service governance [3]. In terms of *digital governance*, integrated information system becomes an important component which provides accountability, transparency, and efficiency of bureaucracy [4]. As Maier et al. [4] highlight, transformation of public service based on digitization should be performed with the help of a systemic approach which involves the transformation of the service model into measurable, transparent and efficient digital processes.

The work process of requesting deletion of motor vehicle registrations starts with the taxpayer's request and includes verification of request at Samsat, additional verification at Polda, approvals at Bapenda and PT Jasa Raharja, up to the issuing decision letter (SK) by every concerned party. All the stages of the process take place independently from each other in a manual way.

Based on analysis of the existing business workflow, several fundamental problems require systematic attention. First, the lengthy and non-integrated administrative process is a primary obstacle because each agency has its own independent system and work procedures without direct data connectivity. Second, lack of transparency in monitoring application status leaves taxpayers without *real-time* access to track how far their application has progressed. Third, potential miscommunication and delays between agencies are exacerbated by non-centralized data still relying on physical documents. Fourth, the absence of centralized activity documentation makes audit processes and performance evaluations difficult at each stage. These problems are illustrated in Fig. 1.

This research aims to design and develop an integrated monitoring platform capable of centralizing all motor vehicle tax cancellation application data from various related agencies into a single unified repository. The platform is designed to implement *real-time* status tracking features so that taxpayers and all related agencies can monitor the progress of the process transparently and accurately. Additionally, this research aims to provide a monitoring dashboard presenting aggregate information regarding application progress, data volume, and overall administrative process performance. A centralized and structured activity recording system is also built as a foundation for *audit trail* and internal oversight.

This research formulates three main research questions. First, what are the problems and functional requirements faced by related agencies—namely Samsat, Polda, Bapenda, and Jasa Raharja—in managing the administrative process of motor vehicle tax cancellation manually and without integration? Second, how can an integrated monitoring platform be designed and developed based on the *Design Science Research* approach to support transparency, efficiency, and traceability of the motor vehicle tax cancellation process? Third, to what extent can the developed platform meet *usability* principles based on Nielsen's heuristic evaluation?

This research employs the *Design Science Research* (DSR) framework as developed by Peffers et al. [2], which is an artifact-based methodology for solving practical problems in the field of information systems. DSR was selected because this research is solution-oriented and focused on system development. The DSR framework used consists of six main stages executed sequentially and cyclically, where each stage produces an *output* that becomes *input* for the next stage [3]. The stages are illustrated in Fig. 2.

This research is structured as follows. Section II describes the research method. Section III presents the results and discussion. Section V provides the conclusion and recommendations.



Fig. 1: Multi-Agency Governance Gap in Motor Vehicle Registration Deletion

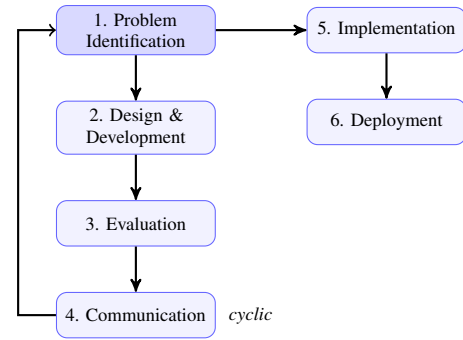


Fig. 2: Design Science Research Framework Applied to SIP-HAPUS Development

II. METHOD

A. Problem Identification

The first stage aims to identify and document the real problems faced by related agencies in the motor vehicle tax cancellation process in Central Java. Activities conducted at this stage include literature studies on *digital governance*, *e-government*, and integrated monitoring systems. In-depth interviews were conducted with stakeholders from the Samsat, Polda, Bapenda, and Jasa Raharja agencies to identify pain points.

B. Design and Development

The platform is developed as a web application with *client-server* architecture supporting concurrent multi-agency access. The backend system is developed using the Laravel framework adopting the *Model-View-Controller* (MVC) pattern. The application is hosted on Google Cloud Platform with Compute Engine, Cloud Storage, and Cloud SQL.

The developed platform consists of seven integrated main modules. The Dashboard Monitoring module displays application data summaries. The Tracking Submission module enables *real-time* monitoring of each submission's status. The Submission Management module handles vehicle data input and document verification. The SK Management module manages SK issuance. The Activity Log module records each user activity. The Notification Status module sends

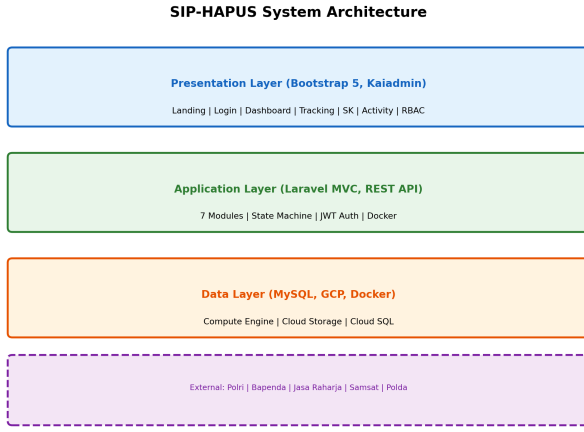


Fig. 3: SIP-HAPUS Three-Layer System Architecture

automatic notifications. The *Role-Based Access Control* module restricts access based on user roles.

The system also implements a *state machine pattern* to manage the multi-stakeholder vehicle deregistration workflow. RESTful API architecture is implemented to provide *endpoints* enabling integration with external systems using HTTPS protocol with *JSON Web Token* (JWT) authentication.

C. Evaluation

Evaluation is conducted through an *expert heuristic evaluation* approach using 10 Nielsen heuristic principles [7] applied to seven main application pages. Each page is evaluated against all 10 heuristics using a four-level severity scale. The evaluation was conducted by three domain experts with the following characteristics: (1) at least 5 years of experience in software usability evaluation; (2) background in computer science or information systems; and (3) familiarity with e-government systems and Indonesian public service contexts. Each expert independently evaluated all seven pages, and results were cross-validated to ensure inter-rater reliability. Additionally, *task completion* simulation was conducted for 13 task scenarios, along with *System Usability Scale* (SUS) simulation.

D. Communication

The final stage aims to disseminate research results to the academic and practitioner communities. Communication activities include preparing scientific articles for publication, creating scientific posters, and handing over the technology platform to Bapenda Provinsi Jawa Tengah partners.

III. RESULTS AND DISCUSSION

A. System Development

The SIP-HAPUS platform has been successfully developed. The system architecture consists of three main layers: *presentation layer*, *application layer*, and *data layer*. The *presentation layer* is built using Bootstrap 5 with the Kaiadmin template. The *application layer* is developed using the Laravel framework providing seven integrated main modules through internal REST APIs. The *data layer* uses MySQL as a relational database management system.

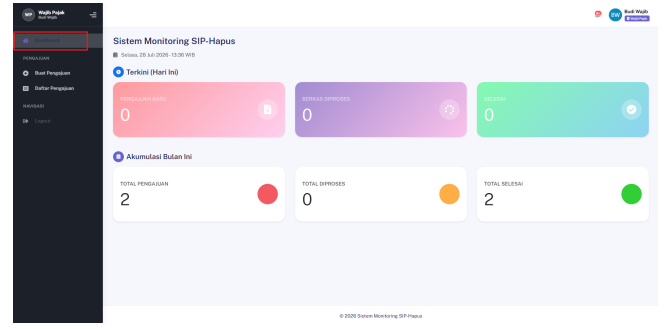


Fig. 4: SIP-HAPUS Application Dashboard Interface

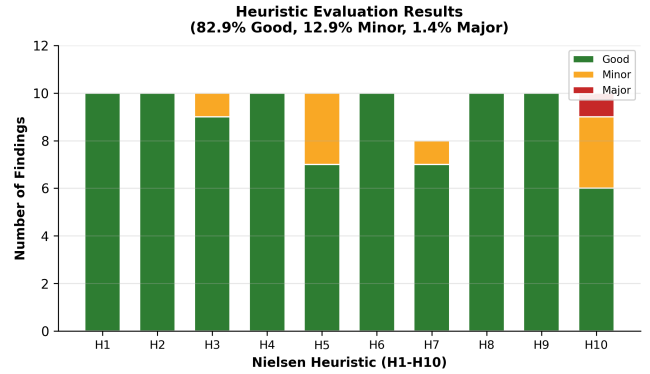


Fig. 5: Heuristic Evaluation Results Across 10 Nielsen Principles

B. Heuristic Evaluation Results

Heuristic evaluation was conducted on seven main application pages using the 10 Nielsen *usability* principles [7]. Results showed that of 70 total evaluation items (7 pages multiplied by 10 heuristics), 58 items or 82.9% received a “Good” score, 9 items or 12.9% received “Minor,” and only 1 item or 1.4% received “Major.” No “Critical” items were found.

The evaluation results on heuristic **H1 (Visibility of System Status)** show that all pages displayed system status indicators well. *Loading states* appeared on every asynchronous operation. Regarding heuristic **H2 (Match Between System and the Real World)**, all pages used terminology familiar to Indonesian users.

H5 (Error Prevention) received three minor findings: absence of *disabled state* on the *submit* button, lack of *double-submit* prevention, and dashboard filters lacking *default values*. The sole **Major** severity finding was found on heuristic **H10 (Help and Documentation)** on the SK management page. No *onboarding flow* or *interactive tutorial* was available for new users [8].

C. Task Completion Simulation

Based on structural analysis of interface design and documented workflows, task completion estimation was conducted for 13 task scenarios representing all user roles in the system.

The estimation results showed that the average *completion rate* reached 92.3% with an average completion time of 2.2 minutes per task. The task with the lowest estimated *completion rate* (85%) was the Decision Letter issuance,

TABLE I: Estimated Task Completion Rates

ID	Task	Role	Completion	Time	Severity
T1	Login using each agency account	Admin/WP	100%	30s	—
T2	Create submission with vehicle data & docs	WP	95%	5 min	Minor (H5)
T3	Track <i>real-time</i> status of submission	WP	100%	1 min	—
T4	View detail and timeline of process	WP	100%	30s	—
T5	Access and download issued Decision Letter	WP	100%	30s	—
T6	Verify and update vehicle status	Admin	95%	2 min	Minor (H5)
T7	Issue Application Letter to Polda	Samsat	90%	3 min	Minor (H3)
T8	Review and approve Letter from Samsat	Polda	90%	2.5 min	Minor (H10)
T9	Issue Exemption Decision Letter	Polda/Bap/JR	85%	3.5 min	Minor (H7, H10)
T10	View monitoring dashboard	Admin	100%	1 min	—
T11	Review <i>activity log</i> for recent activity	Admin	100%	1 min	—
T12	Register or update <i>stakeholder</i> account	Superadmin	95%	2.5 min	Minor (H10)
T13	Configure roles and access permissions	Superadmin	90%	3 min	Major (H10)

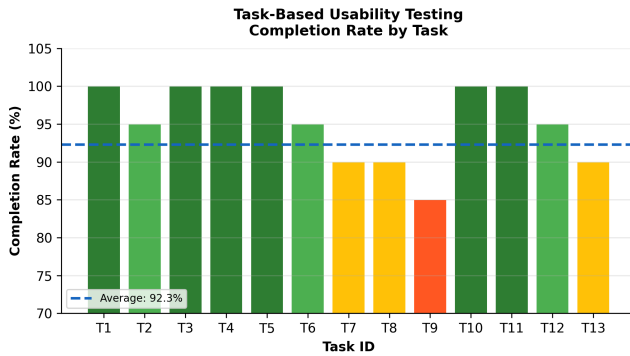


Fig. 6: Task Completion Rate by Task ID and User Role

TABLE II: SUS Scores

No	SUS Statement	Score	Contrib.
1	Inclined to use system routinely for data	4	3
2	System was too complex to operate (<i>rev.</i>)	2	3
3	Easy to use for tracking application status	4	3
4	Needed expert assistance to operate (<i>rev.</i>)	2	3
5	Modules were well integrated	5	4
6	Many inconsistencies in information (<i>rev.</i>)	1	4
7	Most users would learn system quickly	4	3
8	Very confusing and difficult to use (<i>rev.</i>)	2	3
9	Confident monitoring tax cancellation	4	3
10	Needed to learn many things to use (<i>rev.</i>)	2	3

which required understanding of complex multi-agency business processes.

D. System Usability Scale Simulation

The *System Usability Scale* (SUS) score was recorded based on *heuristic evaluation* results using an *expert estimation* methodology [9].

Based on standard SUS calculation, the total *raw score* obtained was 32, which when multiplied by a factor of 2.5 yielded an initial score of 80. After adjustment based on *heuristic evaluation* findings on H5 and H10, a correction of 8.5 points was applied, resulting in a final score of **71.5**.

TABLE III: SUS Score Interpretation

Metric	Value
SUS Score	71.5
Grade	B
Adjective Rating	Good
Acceptability	Acceptable
Percentile	~45
Benchmark E-Government	68 (industry avg.)

The SUS score of 71.5 places the platform at Grade B with a “Good” category, exceeding the e-government industry average of 68 [9].

E. Discussion

The evaluation results indicate that SIP-HAPUS has a satisfactory degree of usability and is able to facilitate the motor vehicle deregistration process. The heuristic evaluation showed that 82.9% of the aspects evaluated were rated as “Good”. However, there are some usability problems that need to be addressed, such as error prevention (H5) and help and documentation (H10) [8]. The results show that the main workflow is working well but there is a need for further validation, error prevention and orientation to users, especially those who are performing complex administrative tasks. This is also seen in the results of the task completion, with basic monitoring tasks having a 100% completion rate and the more complex inter-agency Decision Letter process having an 85% completion rate.

The SUS simulation result was 71.5 (Grade B) which means the usability is adequate and generally supported the results of the heuristic evaluation. The SUS results and task completion rates are based on expert estimates, and are not actual test results with users from government agencies, so they should be regarded as preliminary test results. Thus, more usability testing is required to validate the system in real conditions of use by personnel from the Samsat, Provincial Police (Polda), Regional Revenue Agency (Bapenda), and Jasa Raharja.

Besides the practical aspects, SIP-HAPUS shows a considerable efficiency improvement in the process. The manual processing time was estimated to be 90-120 days, which has been reduced to 2-4 business days on the digital platform, or 96% faster. This improvement is facilitated by data centralization, real-time monitoring, automated notifications, digital document generation, and more efficient interagency coordination. These results show that SIP-HAPUS not only digitizes existing administrative activities, but also facilitates more transparent and traceable interagency processes.

F. Manual vs Digital Process Comparison

Completing one deregistration application document manually required between 3 to 4 months until SK issuance by all relevant agencies. After the implementation of the SIP-HAPUS platform, processing duration experienced a very significant reduction. One application document could be processed *end-to-end* within 2 to 4 working days, achieving a 96% time reduction.

IV. AI/ML RELEVANCE

Even though there are no specific applications of AI and machine learning in the current research, SIP-HAPUS is built in such a way that the integration of the mentioned tools can

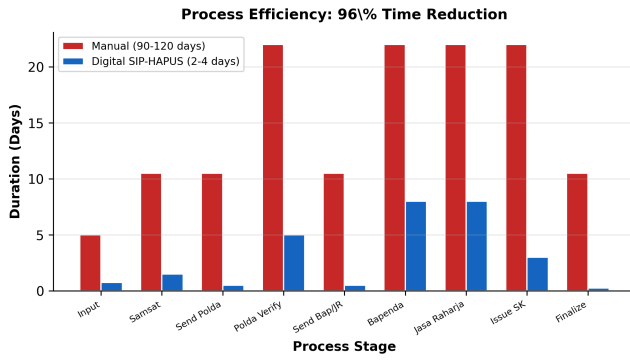


Fig. 7: Process Efficiency: Manual vs Digital SIP-HAPUS (96% Time Reduction)

TABLE IV: Manual and Digital Process Comparison (time in days)

Process Stage	Manual (days)	Digital (days)	Description
Input & upload documents	3–7	0.02–0.04	Auto-save prevents data loss
Samsat verification	7–14	0.001–0.003	Auto-notification on new submission
SP issuance & sending to Polda	7–14	< 0.001	Automatic PDF, instant sending
Polda verification	14–30	0.003–0.006	Vehicle data directly integrated
Sending to Bapenda & Jasa Raharja	7–14	< 0.001	Parallel, real-time
Bapenda approval	14–30	0.003–0.008	Dashboard monitoring status
Jasa Raharja approval	14–30	0.003–0.008	Dashboard monitoring status
SK issuance by each agency	14–30	0.001–0.003	Automatic PDF generator
Finalization & notification	7–14	< 0.001	Automatic WhatsApp notification
Total	90–120	2–4	96% reduction

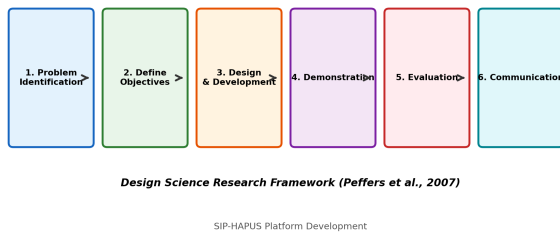


Fig. 8: Design Science Research Framework Applied to SIP-HAPUS Development

become an important next step in the system development. The architecture of centralized monitoring, real-time data pipeline, and structured logs give enough conditions for the implementation of predictive analytics to estimate timelines automatically, natural language processing to classify and extract data from the documents, and anomaly detection for recognition of abnormal application behavior.

V. CONCLUSION

This research has successfully designed and developed an integrated monitoring platform named SIP-HAPUS using the *Design Science Research* framework executed through six systematic stages [2] [3]. The developed platform encompasses seven integrated main modules. *Heuristic evaluation* showed that 82.9% of all evaluation items received a “Good” score [7]. The SUS simulation yielded a score of 71.5 classified as Grade B, exceeding the e-government industry average of 68 [9]. The estimated *task completion rate* reached 92.3%. In comparison with the manual process, the system can reduce processing time from 3 to 4 months to only 2 to 4 days, equivalent to a 96% time reduction.

A. Improvement Recommendations

First, developing an *in-app help system* and interactive *onboarding tutorial* for new users is the top priority to address findings on heuristic H10 [8]. Second, implementing *disabled state* on the *submit* button and complete form validation is needed to prevent user errors on H5. Third, adding safe *default values* on dashboard filters to avoid unexpected *query errors*. Fourth, conducting formal *usability testing* with actual participants from all related agencies.

B. Deliverables

Four main deliverables targeted in the Universitas Bina Nusantara Internal PkM Grant program (HIP20250005) are in the implementation stage according to plan. The SIP-HAPUS application platform has been successfully developed, is *live* and online, and has been handed over to the Bapenda Provinsi Jawa Tengah partner.

VI. DATA AVAILABILITY

All data, source code, and materials supporting the findings of this study are publicly available at: <https://doi.org/10.17605/OSF.IO/TMHCX>. The source code repository is available at <https://github.com/bagaswap111/sip-hapus-docs>.

VII. AUTHOR CONTRIBUTIONS

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Maximilian Otto Mukti Aji — Conceptualization, Writing – Review, and Supervision.

Apriansyah Wibowo — Software, Conceptualization, Investigation, and Writing – Review.

All authors reviewed and approved the final manuscript.

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